16. Implement a Sobel algorithm using Open CV to filter the input image.

Code

```
import cv2
import matplotlib.pyplot as plt

# Read the input image
img = cv2.imread('image.jpg')

# Check whether the image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Convert the image to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Apply Sobel filter in the X direction
    sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

    # Apply Sobel filter in the Y direction
    sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

    # Convert results to 8-bit images
    sobel_x = cv2.convertScaleAbs(sobel_x)
    sobel_y = cv2.convertScaleAbs(sobel_y)

    # Combine both directions
    sobel_combined = cv2.addWeighted(
        sobel_x, 0.5, sobel_y, 0.5, 0
    )

    # Display the images
    plt.figure(figsize=(10, 8))

    plt.subplot(2, 2, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title("Original Image")
    plt.axis("off")

    plt.subplot(2, 2, 2)
    plt.imshow(gray, cmap='gray')
    plt.title("Grayscale Image")
    plt.axis("off")

    plt.subplot(2, 2, 3)
    plt.imshow(sobel_x, cmap='gray')
    plt.title("Sobel X")
    plt.axis("off")
```

```
plt.subplot(2, 2, 4)
plt.imshow(sobel_combined, cmap='gray')
plt.title("Sobel Edge Detection")
plt.axis("off")

plt.tight_layout()
plt.show()
```

